Machine Learning

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Lecture 26: Unsupervised Learning

Supervised Learning

$$D = \{\left(\overrightarrow{x^{(1)}}, y^{(1)}\right), \left(\overrightarrow{x^{(2)}}, y^{(2)}\right), \ldots \left(\overrightarrow{x^{(m)}}, y^{(m)}\right)\} \subseteq X \times Y$$

Where,

- D is the data set
- X is the d-dimensional feature space ($\mathbb{R}^d$)
- $\overrightarrow{x^{(i)}}$ is the input vector of the ith sample
- Y is the label space

The data points are drawn from an unknown distribution P

$$\left(\overrightarrow{x^{(i)}}, y^{(i)}\right) \sim P(x, y)$$

We want to learn a function $h \in H$, such that for a new instance $(x, y) \sim P$,

$$h(x) = y$$

With a high probability or at least, $h(x) \approx y$.

Unsupervised Learning

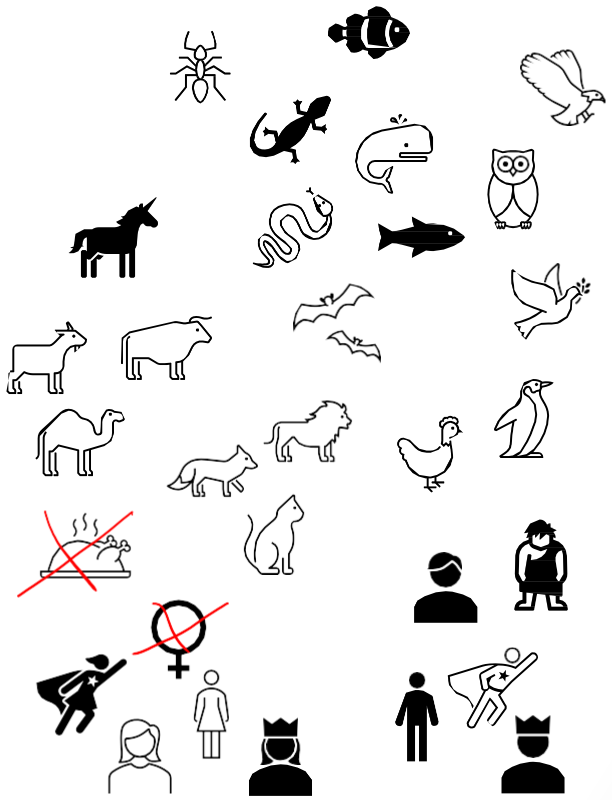
$$D = \{\overrightarrow{x^{(1)}}, \overrightarrow{x^{(2)}}, \overrightarrow{x^{(3)}}, \dots, \overrightarrow{x^{(m)}}\} \subseteq X$$

Where,

- D is the data set
- X is the d-dimensional feature space ($\mathbb{R}^d$)
- $\overrightarrow{x^{(i)}}$ is the input vector of the ith sample
- No labels!

Questions:

- Find the underlying structure of the data or patterns in the data
  - Not about prediction
- What subpopulations exist in the data?
  - How many? And how big? How cohesive? Outliers?
  - Do elements in each subpopulation have commonalities?
  - Is there a hierarchy?



Use cases of Unsupervised learning

- **Customer segmentation:** Understanding different customer groups around which to build marketing or other business strategies (e.g., needs exercise equipment based on age and gender)
- **Social media analysis:** Various types of users, how they interact with one another, the information they contribute and consume, etc. (e.g., which groups of users generate misinformation?)
- **Genetics:** Clustering DNA patterns to analyze evolutionary biology
- **Astronomical data analysis:** Recognize similar star and galactic clusters
- **Organize computing clusters:** Reorganize and optimize resources based on patterns of utilization and working together
- **Recommender systems**: Grouping together users with similar viewing patterns in order to recommend similar content (customers who viewed x also viewed y)
- **Anomaly detection**: e.g., fraud detection or detecting defective mechanical parts (predictive maintenance).
- **Feature selection**: e.g., dimensionality reduction

Types of clustering

- **Goal of the clustering**
  - **Monothetic:** Cluster members have some common property
    - Uses the features one by one
    - E.g., all males have ages between 20 and 30 years, all patients of type A have y% response to drug B
  - **Polythetic:** Cluster members are “similar” to each other but no single common property
    - Uses the features all at once
    - E.g., distance between elements defines membership
- **Overlap:**
  - **Hard clustering:** Clusters do not overlap, and each element belongs to one or the other
  - **Soft clustering:** Clusters may overlap, “Strength” of association between element and cluster
- **Flat vs. hierarchical**
  - Sets of groups vs. taxonomy

Methods

- **K-D Trees**
  - Monothetic, hard boundary, hierarchical
- **K-Means**
  - Splits the data into predefined number (K) of clusters
  - Polythetic, hard boundary, flat
- **Gaussian Mixture Models**
  - Fits a mixture of K gaussians to the data
  - Polythetic, soft boundary, flat
- **Agglomerative clustering**
  - Creates and ontology of nested sub-populations
  - Polythetic, hard boundary, hierarchical

Thank you!